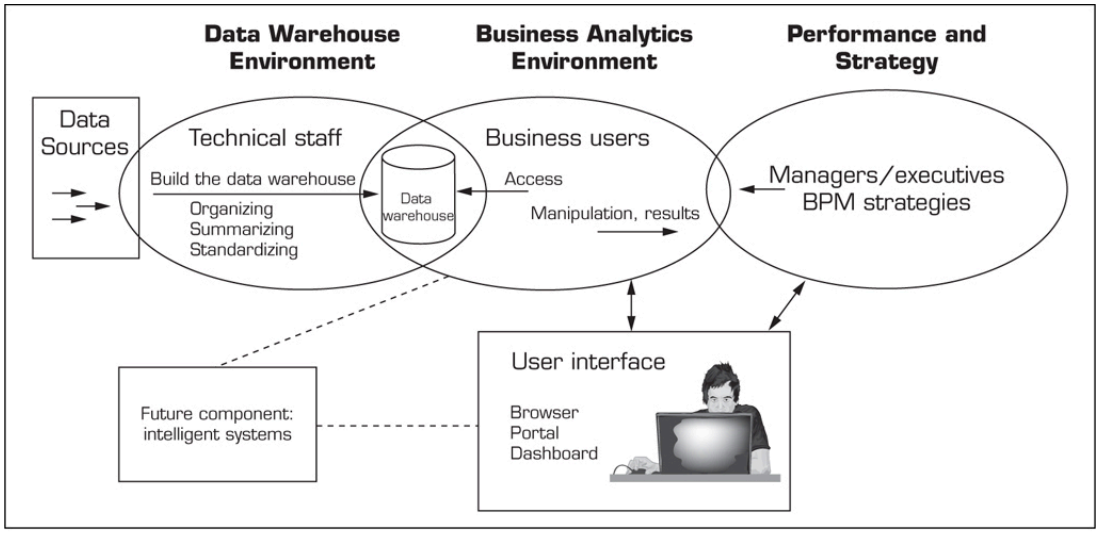
ISA 401 Notes

Lecture 1: Introduction to BI and Data Viz - 8/24/26 and 8/26/26

- ChatISA - can use and compare many models
- Unit of analysis, what does a row of data mean?
    - In today's class, a unit of analysis is a person involved in a crash, not just a crash. So, multiple people = multiple rows for 1 crash
    - Even though Friday had the most crashes, higher volume of drivers means that Friday isn't necessarily less safe to drive
        - Correlation ain't causation peeps - tsts
        - Encoding counties, means how do you decide what is a county - so just x and y position, not color of the graph
        -
- **Pre-Analytics/Data Management**: extracting the needed data for analysis. Data can either be:
    - Stale, convenient, archived
    - Fresh, challenging, impactful
- **Descriptive Analytics -** where one attempts to understand the data through descriptive statistics and visualizations
- **Predictive Analytics** - what will happen?
    - Statistical/ML models trained on historical data
- **Prescriptive Analytics -** what should we do?
    - How do you use your resources to reduce crashes
    - Suggest actions, not just forecasts
- **What is Business Intelligence? Data -> Decisions**
    - "... to enable **interactive access (sometimes in real time)** to data, to enable manipulation of data, and to give business managers and analysts the ability to conduct appropriate analysis. By analyzing ... data, situations, and performances, decision makers get valuable insights that enable them to **make more informed and better decisions** ... BI is based on **the transformation of data to information, then to decisions, and finally to actions**."
    - Business Intelligence: Pre-analytics and descriptive stages
- The BI Process (pic down below)



Source: Based on W. Eckerson, Smart Companies in the 21st Century: The Secrets of Creating Successful Business Intelligent Solutions. The Data Warehousing Institute, Seattle, WA 2003, p. 32, Illustration 5.

- **Unstructured Data** - free text with no fixed fields; text, images, audio
    - Find a way to convert that data into something more insightful
- **Semi-Structured:** carries its own labels but records can differ
- **Structured Data:** rows * columns with identical fields throughout (tables)
- **Data Visualization** - involves presenting data in a graphical format. It is really a process that starts by getting data, creating initial plot(s) and modifying them to answer questions of interest. A good chart takes time and iteration
- Visual Encoding - what techniques do you use to help convey the insights from the data (color, sorting order, size)
    - What are your variables and how are you putting these variables in the visualization?

Lecture 2: (R foundations and Reproducibility

- Why use scripts → spreadsheets can have errors, and this errors can create a lot of issues and be costly to an organization
- If you use a spreadsheet, reconfiguring the data around a new unit of analysis is complicated
- Scripts allow you to understand what someone else did, and is more flexible in the event that a change is needed to be made
- Code is just text, and there are many benefits of its use
    - **Readable -** you can read what someone else did, top to bottom
    - **Open -** an error message is text too, search it or paste it into AI
    - **Reusable -** new month, new file: change 1 line and re-run; someone else gets the same result

    - **Diffable -** text can be compared line by line, so a tool can show you exactly what changed, and when. That tool is next.
- Git - a program on your computer that keeps snapshots of your folder (Git is the tool and it runs on your machine and works offline)
- GitHub - a website that hosts Git folders so people can share, review, and publish them (GitHub is the place; copies live there so they can be shared, reviewed, and published)
- **Git's Terminology**
    - Repository (repo) - a folder Git watches
    - Commit - a named snapshot of that folder (only on that specific computer)
    - Push/Pull - upload/download snapshots to or from GitHub.com
    - Branch - a parallel copy for work in progress
    - Pull Request - "Please merge my branch", plus discussion
    - Merge - accept it into main
- Git can show specific changes in text files (**Code, CSV, Markdown, .R, .Rmd, .qmd, HTML**
- For binary files(**.xlsx, .pptx, .docx, .png, .pdf** , doesn't show specific changes, only new file version
- When you edit and save a README.md, then when you click "Commit to main" the snapshot is only on the computer used
- Pushing means you are pushing it to the web (server)
- Pulling means that you are allowing the commits to be on different computers
- Revert: new commit that reverses old commit
- File types:
    - R and HTML files show what changed on a line by line basis
    - PDF, pngs, etc. show…
- You pushed a commit with a mistake in it, the fix that keeps the history honest is Revert, which adds a new commit that reverses the old one

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Lecture 3: R Foundations

- Project: business_intelligence repo
- See HTML slides for global settings changes
- Rscript: just code
- Rmarkdown: text, output, etc.
- Lines 1-5 called YAML
    - Get another markdown language
- Toc: table of contents, adds to top of markdown page
- toc_float: true, now on left side
- Number_sections: true
- Code_folding: show - makes it so you can hide or show code chunks in rmd
- code_download: true
- ```{r NAME, include=FALSE}

knitr::opts_chunk$set(echo = TRUE) - echo true is showing code, echo false is hiding code

- **Three parts**
- **Name** on the left: crashes_per_day. Letters, digits, _ and .; no spaces; case matters.
- **Assignment** in the middle: =, <-, or ->. We write =. You will see <- in most R code online and in my editor, and -> mostly at the end of a pipeline. All three do the same job.
- **Value** on the right: whatever the expression works out to

- Errors
    - Object not found: capitalization, typo, or never ran chunk

- `c()` **combines** values into a **vector**: one type, in order, optionally with names.
- An operation on a vector runs on **every element**: `> 2000` returned seven answers, not one.
- `sum()` of seven TRUE/FALSE values counts the TRUEs.

- Types of vectors
    - Logical: true/ false
    - Integer: whole numbers (written with an L, ex. 2026L)
    - Double: any number
    - Character: text
    - Complex: imaginary numbers
    - Raw: bytes

```
crashes_by_day_int = c(2020L, 2241L, 2327L, 2285L,
                       2380L, 1866L, 1664L)
```{r type_of_vec}
typeof(crashes_by_day_int)
```
```

Lecture 4: Data Import and Export

- An atomic vector is a one-dimensional column in a data table
- Data Frames - each column is one vector with one type (<chr>,<dbl>,<lgl>)
    - All columns have the same length, so every row lines up
- Tibble - the tidyverse's data frame: same object, tidier printing
    - Process of moving from a one-dimensional columns to an organized set of columns and rows
    - The only way to move from a vector to a data frame is that the vectors have to be the same length
- A matrix can only have one data type and one column name (not as useful for analysis)

- Glimpse function gives you a look at the types of data, the data, and the names of the columns
    - Equivalent of str() but prints neater
- List - allows you to add vectors that may not be the same size; a list holds anything: vectors of different types and lengths, and even other lengths
-

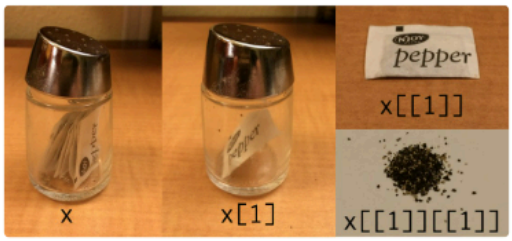
[ ] vs [[ ]] on a List

```
str(example_list[1])     # [ ]  the shaker: a list of 1
List of 1
 $ : int [1:3] 1 2 3
```

```
str(example_list[[1]])   # [[ ]] the packet: the element
 int [1:3] 1 2 3
```

```
example_list[[1]][[1]]   # the pepper: first value in it
[1] 1
```

```
example_list[[4]][2]     # element 4, then its 2nd value
[1] 5.9
```



The pepper shaker rule

`x` is the shaker. `x[1]` is a shaker with one packet: **still a list**. `x[[1]]` is the packet itself. `x[[1]][[1]]` is the pepper.

When the list has **names**, `x$name` is `x[["name"]]` with less typing. Next slide.

- $ lets you pull specific elements
- When subsetting, there are 3 things that can go inside [ ]
    - One position, counting from 1, not 0
    - A name, when the vector has names
    - A condition: a TRUE/FALSE vector as long as the original keeps the positions where it is TRUE.
- A negative position such as [-1] drops instead of keeps
- $ and [[ ]] open one column into a vector. [rows, columns] hands back a smaller tibble; leave a side of the comma empty to keep all of it.

Lecture 5: Data Import, SQLite, and First Deployed App

```
> con = DBI::dbConnect(RSQLite::SQLite(), "../data/scout.db")
> rm(con)
> con = DBI::dbConnect(RSQLite::SQLite(), "../data/scout.db")
> DBI::dbListTables(con)
character(0)
> DBI::dbDisconnect(con)
> con = DBI::dbConnect(RSQLite::SQLite(), "../data/scout.db")
```

```
> DBI::dbListTables(con)
[1] "scout_postings" "scout_runs"
> postings = DBI::dbReadTable(con, "scout_postings")
> DBI::dbDisconnect(con)
> dplyr::glimpse(postings[, c("title", "company",
+  "location_state", "remote", "skills_json")])
Rows: 1,891
Columns: 5
$ title          <chr> "Data Analyst - Remote", "Business
Analyst / Product Owner", "Business Analyst 2 - Enterprise
Application Support", "Sa…
$ company        <chr> "UnitedHealth Group", "Anza Mortgage
Insurance Corporation", "CoServ", "Canva", "Canva", "GFL
Environmental Inc.", "Uli…
$ location_state <chr> "GA", "NC", "TX", "CA", "TX", "NC", "IL",
"GA", "IN", "CA", "TX", "NC", "CA", "IL", "PA", "OH", "IL",
"TX", "NC", "NY",…
$ remote         <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
0, 0, 0, 0, 0,…
$ skills_json    <chr>
"[{\"skillId\":\"sql\",\"importance\":\"required\"},{\"skillId\"
:\"excel\",\"importance\":\"required\"},{\"skillId\":\"…
```

```
ft_remote = postings |>
  dplyr::group_by(category,location_state) |>
  dplyr::summarise(postings = dplyr::n(), remote_share = mean(remote),
                   .groups = "drop"
  )
ft_remote
```